

AZURE DATA ENGINEERING JOURNEY

String Functions

13 string functions, one Superstore dataset — hands-on SQL practice notes from today's session.

SWIPE TO LEARN →

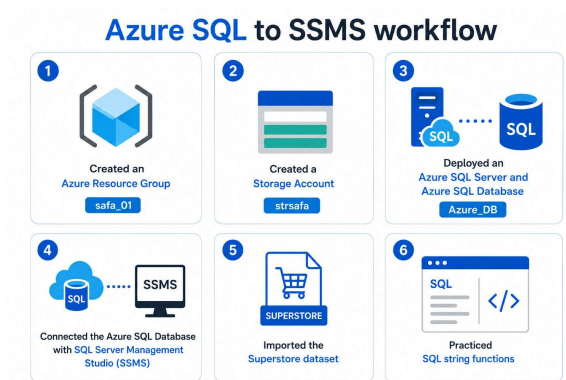
CLOUD SETUP · AZURE

Cloud-to-Desktop Workflow

02

Provisioned the Azure infrastructure end-to-end, then connected it locally in SSMS — taking a project from an empty resource group to a live, query-ready database.

- ✓ Resource Group created
- ✓ Storage Account provisioned
- ✓ Azure SQL Server + Database deployed
- ✓ Connected to Azure SQL via SQL Server Management Studio (SSMS)
- ✓ Superstore dataset imported and query-ready



CODE SNIPPET

 SSMS Practice

03

Using the Superstore Dataset, practiced the string functions on SSMS.

```
--STRING FUNCTIONS

--CONCAT ADDS TWO OR MORE STRINGS TOGETHER
SELECT [CUSTOMER NAME], [COUNTRY/REGION] FROM ORDERS

SELECT [CUSTOMER NAME], [COUNTRY/REGION],CONCAT([CUSTOMER NAME], ' is from ', [COUNTRY/REGION], ' travelled in ',[SHIP MODE]) as NAME_COUNTRY FROM ORDERS

--WSCONCAT ADDS TWO OR MORE STRINGS TOGETHER WITH A SEPARATOR
SELECT [CUSTOMER NAME], [COUNTRY/REGION],CONCAT_WS(' is from ', [CUSTOMER NAME],[COUNTRY/REGION]) AS NAME_COUNTRY FROM ORDERS

--LEFT FUNCTION GIVES THE NUMBER OF LETTERS FROM LEFT
SELECT [CUSTOMER NAME], LEFT([CUSTOMER NAME],5) AS FIRST_4 FROM ORDERS

--RIGHT FUNCTION GIVES THE NUMBER OF LETTERS FROM RIGHT
SELECT [CUSTOMER NAME], RIGHT([CUSTOMER NAME],5) AS LAST_4 FROM ORDERS

--SUBSTRING FUNCTION GIVES THE NUMBER OF LETTERS IN THE MIDDLE
SELECT [CUSTOMER NAME], SUBSTRING([CUSTOMER NAME],5) AS MIDDLE_4 FROM ORDERS

--LENGTH GIVES THE LENGTH OF CHARACTERS
SELECT [CUSTOMER NAME], LEN([CUSTOMER NAME]) AS CHARACTER_COUNT FROM ORDERS

--LOWER FUNCTIONS CONVERTS THE STRING IN LOWER CASE
SELECT [CUSTOMER NAME], LOWER([CUSTOMER NAME]) AS LC FROM ORDERS

--UPPER FUNCTIONS CONVERTS THE STRING IN UPPER CASE
SELECT [CUSTOMER NAME], UPPER([CUSTOMER NAME]) AS UC FROM ORDERS

--LTRIM FUNCTION REMOVES THE SPACES FROM LEFT
SELECT LTRIM('    SAFA    ') AS LTRIM

--RTRIM FUNCTION REMOVES THE SPACES FROM RIGHT
SELECT RTRIM('    SAFA    ') AS RTRIM

--TRIM FUNCTION REMOVES THE SPACES BOTH LEFT AND RIGHT
SELECT TRIM('    SAFA    ') AS TRIM

--REPLACE IS USED TO REPLACE ALL OCCURRENCES OF A SUBSTRING WITHIN A STRING WITH A NEW SUBSTRING
SELECT REGION, REPLACE(REGION,'CENTRAL','NORTH CENTRAL') AS REPLACE FROM ORDERS

SELECT REPLACE ('SQL CLASS', 'C','G')

SELECT PATINDEX ('%KAIZEN%', 'WWW.KAIZENKNOWLEDGE.ORG')

--PATINDEX GIVES THE INDEX OF THAT PARTICULAR CHARACTER
--%D--CHARACTER CAN BE ANYWHERE
--D% -- IT WILL IDENTIFY THE LETTER STARTING WITH D
--%D -- IT WILL IDENTIFY THE LETTER ENDING WITH D

SELECT [CUSTOMER NAME], PATINDEX('%D%', [CUSTOMER NAME]) AS CUS_NAME_WITH_D FROM ORDERS

SELECT [CUSTOMER NAME], PATINDEX('D%', [CUSTOMER NAME]) AS CUS_NAME_WITH_D FROM ORDERS

SELECT [CUSTOMER NAME], PATINDEX('%D', [CUSTOMER NAME]) AS CUS_NAME_WITH_D FROM ORDERS

SELECT [CUSTOMER NAME],
    PATINDEX('%D%', [CUSTOMER NAME]) AS D_ANYWHERE,
    PATINDEX('D%', [CUSTOMER NAME]) AS STARTING_WITH_D,
    PATINDEX('%D', [CUSTOMER NAME]) AS ENDING_WITH_D
FROM ORDERS
```

STRING FUNCTIONS · COMBINING TEXT

04

CONCAT vs CONCAT_WS

Two ways to merge text straight out of the ORDERS table — one glues strings together manually, the other applies a separator automatically.

🔗 CONCAT

Adds two or more strings together directly. Great for building a readable, human-friendly label straight from raw columns — raw columns — no separate formatting step required.

```
-- CONCAT adds two or more strings together
SELECT [CUSTOMER NAME], [COUNTRY/REGION] FROM ORDERS

SELECT [CUSTOMER NAME], [COUNTRY/REGION],
       CONCAT([CUSTOMER NAME], ' is from ', [COUNTRY/REGION],
             ' travelled in ', [SHIP MODE]) AS NAME_COUNTRY
FROM ORDERS
```

⋮ CONCAT_WS

Adds two or more strings together with a separator — defined once, instead of retyping it between every single value.

```
-- CONCAT_WS adds strings together with a separator
SELECT [CUSTOMER NAME], [COUNTRY/REGION],
       CONCAT_WS(' is from ', [CUSTOMER NAME],[COUNTRY/REGION])
       AS NAME_COUNTRY
FROM ORDERS
```

STRING FUNCTIONS · SLICE & MEASURE

05

Slicing & Measuring Text

Four functions for working with parts of a string — grab characters from either end, pull from the middle, or measure how long a value is.

<< LEFT

Pulls a set number of characters from the start of a string — string — useful for building short codes or prefixes straight from a name column.

```
SELECT [CUSTOMER NAME],  
       LEFT([CUSTOMER NAME],5) AS FIRST_4  
FROM ORDERS
```

>> RIGHT

Pulls a set number of characters from the end of a string — string — handy for suffixes or the last few digits of an identifier.

```
SELECT [CUSTOMER NAME],  
       RIGHT([CUSTOMER NAME],5) AS LAST_4  
FROM ORDERS
```

⌞ SUBSTRING

Pulls characters from anywhere inside a string once you know the starting position — for when what you need isn't at either edge.

```
SELECT [CUSTOMER NAME],  
       SUBSTRING([CUSTOMER NAME],5)  
       AS MIDDLE_4  
FROM ORDERS
```

LEN

Counts the total number of characters in a string — a quick way to spot unusually short or long values before cleaning a column.

```
SELECT [CUSTOMER NAME],  
       LEN([CUSTOMER NAME])  
       AS CHARACTER_COUNT  
FROM ORDERS
```

STRING FUNCTIONS · CASE CONVERSION

06

UPPER vs LOWER

Two mirror-image functions for standardizing text case — pick whichever case your report, dashboard, or downstream downstream system expects.



UPPER

Converts every letter in a string to uppercase. Common for codes, categories, and any field that needs to look consistent across every row.

```
SELECT [CUSTOMER NAME],  
       UPPER([CUSTOMER NAME]) AS UC  
FROM ORDERS
```



LOWER

Converts every letter in a string to lowercase. Useful for case-insensitive comparisons, lookups, and matching emails or usernames.

```
SELECT [CUSTOMER NAME],  
       LOWER([CUSTOMER NAME]) AS LC  
FROM ORDERS
```

💡 UPPER and LOWER never change the underlying data — they only reshape the query output, so the original column stays exactly as imported.

STRING FUNCTIONS · CLEANING WHITESPACE

07

LTRIM, RTRIM & TRIM

Three ways to strip stray spaces from imported or manually typed text — pick the side (or both) that actually needs cleaning.



LTRIM

Removes spaces from the left side of side of a string only, leaving any trailing spaces on the right untouched.

```
SELECT LTRIM(  
    '    SAFA    '  
) AS LTRIM
```



RTRIM

Removes spaces from the right side of side of a string only, leaving any leading spaces on the left untouched.

```
SELECT RTRIM(  
    '    SAFA    '  
) AS RTRIM
```



TRIM

Removes spaces from both the left and right sides in one step — the safest default for messy, inconsistently typed text.

```
SELECT TRIM(  
    '    SAFA    '  
) AS TRIM
```

💡 Leading and trailing spaces are invisible in query results but can silently break joins and WHERE-clause matches — trimming is a cheap safeguard before comparing text.

STRING FUNCTIONS · FIND & REPLACE

 REPLACE

08

Swaps every occurrence of a substring with a new one, anywhere it appears — a one-line fix for inconsistent formatting, typos, or renamed values across an entire column.

```
-- REPLACE swaps every occurrence of a substring
SELECT REGION, REPLACE(REGION, 'CENTRAL', 'NORTH CENTRAL')
       AS REPLACE
FROM ORDERS

SELECT REPLACE ('SQL CLASS', 'C', 'G')
```

SYNTAX: `REPLACE(string_expression, old_substring, new_substring)`

The second example runs REPLACE straight on a literal string, not a table column — a fast way to sanity-check how a function behaves before pointing it at real data.

COMMON USE CASES

Standardizing region or category names

Stripping unwanted symbols from text

Fixing inconsistent spelling or formatting

STRING FUNCTIONS · PATTERN MATCHING

09

 PATINDEX

Returns the starting position of a pattern inside a string — or 0 when it isn't found. Where you place the % wildcard controls what counts as a match.

%D%

the letter can be anywhere in the string

D%

identifies text starting with D

%D

identifies text ending with D

```
-- test a pattern against a literal string first
SELECT PATINDEX ('%KAIZEN%', 'WWW.KAIZENKNOWLEDGE.ORG')
```

```
-- all three placements in one real query
SELECT [CUSTOMER NAME],
       PATINDEX('%D%', [CUSTOMER NAME]) AS NAME_WITH_D,
       PATINDEX('D%', [CUSTOMER NAME]) AS STARTING_WITH_D,
       PATINDEX('%D', [CUSTOMER NAME]) AS ENDING_WITH_D
FROM ORDERS
```

💡 PATINDEX returns 0 when nothing matches instead of throwing an error — which makes it easy to drop straight into a WHERE or CASE statement to flag records.

QUICK RECAP

STRING TOOLKIT

COMBINE

→ CONCAT · CONCAT_WS

SLICE & MEASURE

→ LEFT · RIGHT · SUBSTRING · LEN

CASE

→ UPPER · LOWER

TRIM

→ LTRIM · RTRIM · TRIM

FIND & FIX

→ REPLACE

PATTERN MATCH

→ PATINDEX

Following my Azure Data Engineering journey — one query at a time.

💡 Let's connect, and feel free to repost if this helped your revision too!